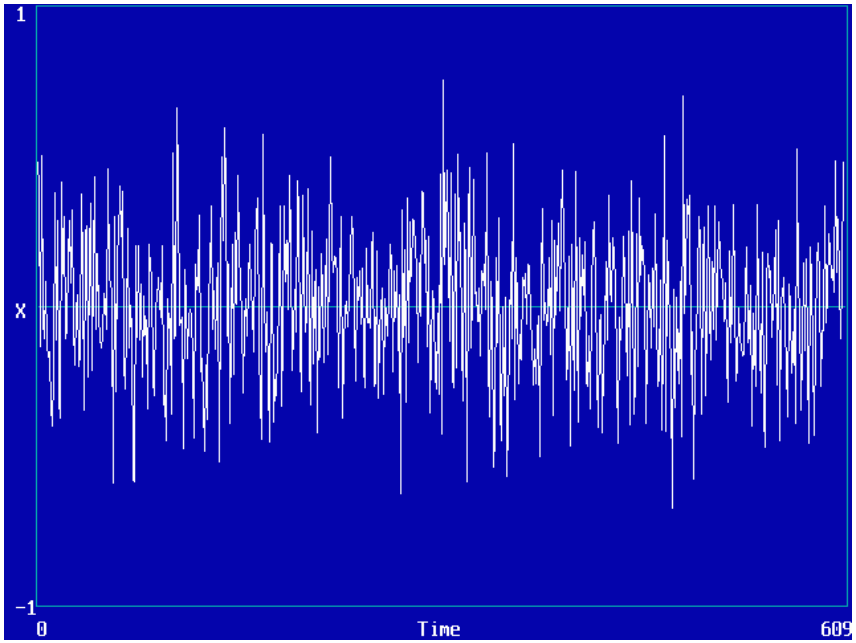
Instrumentation Amplifiers

Sensor Signals

- Most signals from sensors are:
  - Small (~mV)
  - Contain unwanted common mode signals which include:
    - Noise
    - DC Offsets
    - Unwanted frequency components



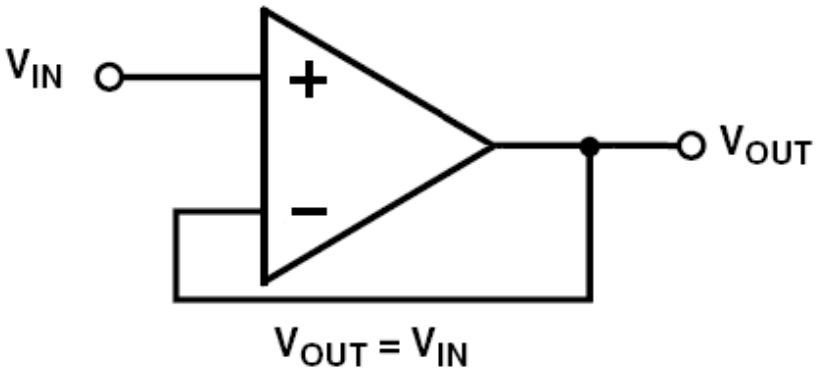
Signal Interfacing

- When interfacing a signal to an analogue to digital converter it is important to remember the following:
  - Buffering
  - Amplification
  - Filtering



Buffers

- Has high input impedance
- Output voltage follows input voltage
- Provides some form of isolation between sensor and following circuit
- Typical buffers include:
  - Voltage Followers
  - Non-inverting Amplifiers



Amplification

- As sensor signals are very small they often need to be amplified
- The signal is multiplied by a gain factor (***A***)

$$V_{out} = A.V_{in}$$

Filtering

- Necessary to filter signal to ensure that is:
  - Nyquist Criterion is satisfied
  - Improve Signal to noise Ratio by reducing noise frequencies
  - Filter out unwanted frequency components such as 50Hz from mains

Common – Mode Rejection (CMR)

- A common mode signal is a signal which appears at both inputs of a device
- CMR is the property of a device which cancels out any signals that are common between both inputs while amplifying signals that are differential between both inputs
- The common mode rejection ratio is the ratio of the differential gain (A_D) to the common-mode gain

$$CMRR = A_D\left(\frac{V_{CM}}{V_{OUT}}\right)$$

- From this we can define the common mode rejection as

$$CMR = 20\log_{10}(CMRR)$$

Instrumentation Amplifier

- A very common and useful concept in instrumentation design
- It amplifies differential signals while rejecting common-mode signals
- It can be made using discrete op amps or available as a monolithic chip
- It is a closed loop gain block that has differential inputs and a single output

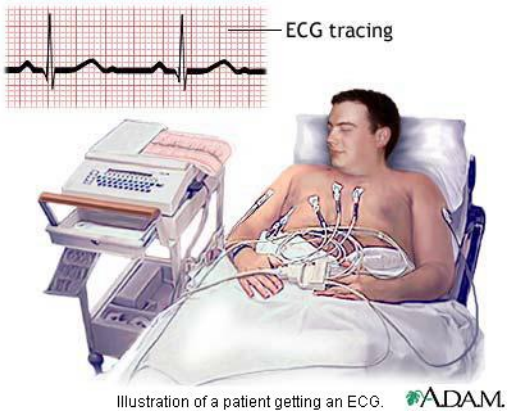


Properties of In Amps

- High AC and DC Common-Mode rejection
- Low offset voltage and offset voltage drift
- Matched, high input impedance (~10^9 Ω)
- Low input bias current (1nA - 50nA)
- Low noise
- Low non linearity
- Simple gain selection
- Good bandwidth
- Rail to rail output voltage swing

Uses of In Amps

- Data acquisition
- Medical instrumentation
- Monitor and control electronics
- Signal conditioning
- Audio applications
- Video applications

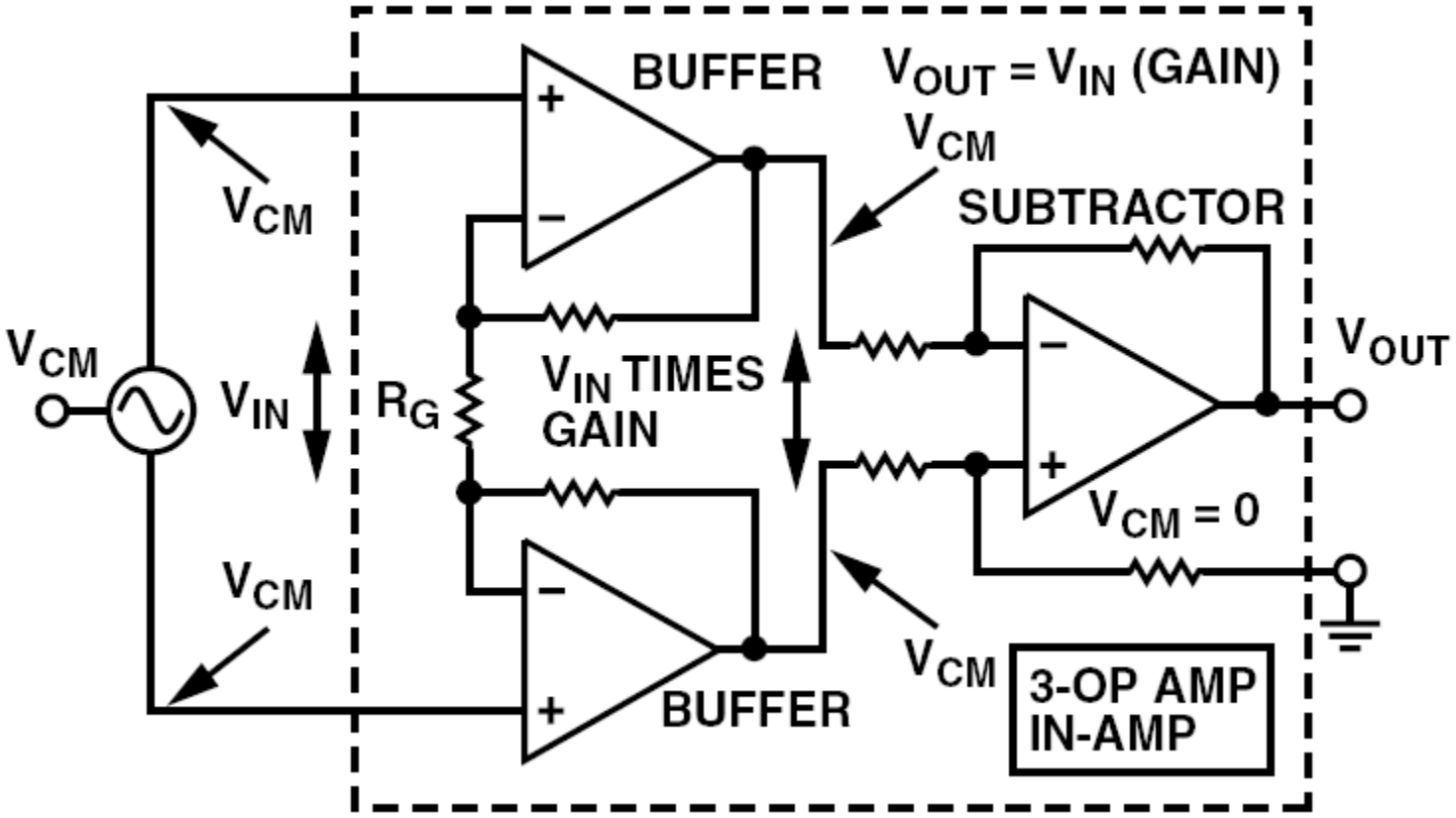






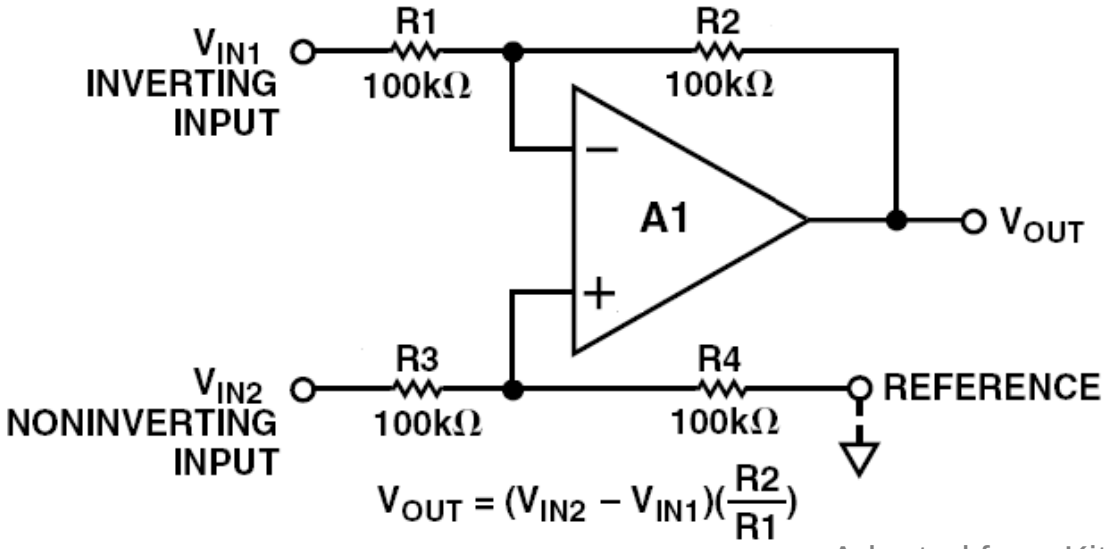


Instrumentation Amplifier Diagram



Taken from Kitchin, C. & Counts, L. (2006)

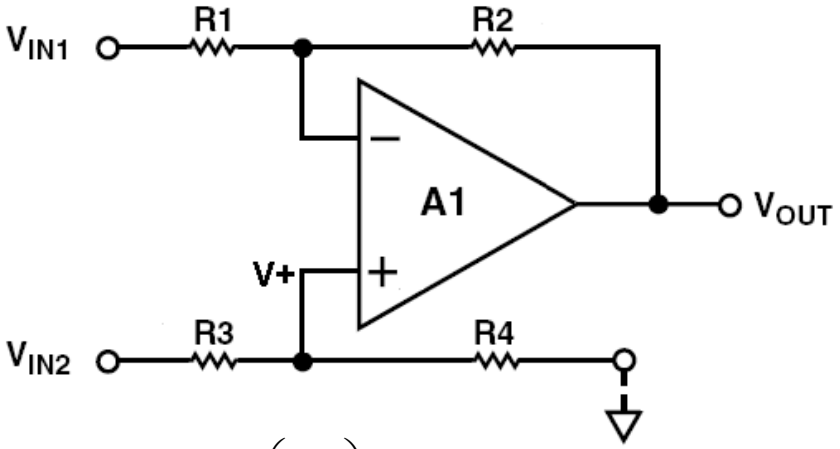
Subtractor



Adapted from Kitchin, C. & Counts, L. (2006)

- It amplifies differential signals while rejecting common-mode signals
- Problems include unmatched input resistances

Subtractor Proof



$$V_{OUT1} = -V_{IN1} \cdot \left(\frac{R_2}{R_1} \right)$$

$$V_{OUT2} = A.V_+ \qquad where \quad V_+ = V_{IN2} \left(\frac{R_4}{R_3 + R_4} \right)$$

$$= A.V_{IN2} \left(\frac{R_4}{R_3 + R_4} \right)$$

$$= V_{IN2} \left(\frac{R_4}{R_3 + R_4} \right) \left(\frac{R_1 + R_2}{R_1} \right)$$

$$V_{OUT} = V_{OUT1} + V_{OUT2}$$

$$= V_{IN2} \left(\frac{R_4}{R_3 + R_4} \right) \left(\frac{R_1 + R_2}{R_1} \right) - V_{IN1} \cdot \left(\frac{R_2}{R_1} \right)$$

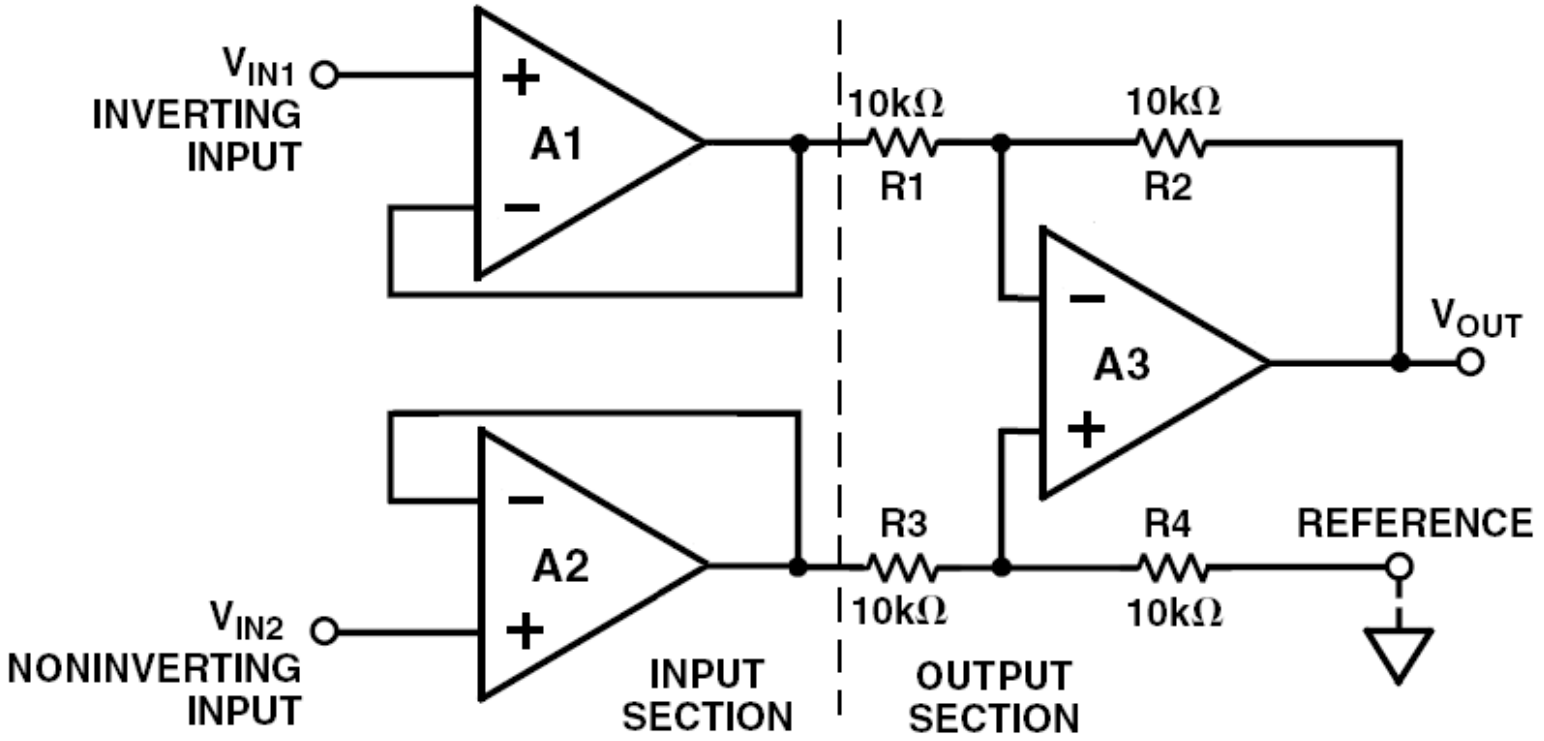
if $R_1 = R_3$ *and* $R_2 = R_4$ *then*

$$V_{OUT} = V_{IN2} \left(\frac{R_2}{R_1 + R_2} \right) \left(\frac{R_1 + R_2}{R_1} \right) - V_{IN1} \cdot \left(\frac{R_2}{R_1} \right)$$

$$= V_{IN2} \left(\frac{R_2}{R_1} \right) - V_{IN1} \cdot \left(\frac{R_2}{R_1} \right)$$

$$= (V_{IN2} - V_{IN1}) \left(\frac{R_2}{R_1} \right)$$

3 Op Amp In Amp with Buffering

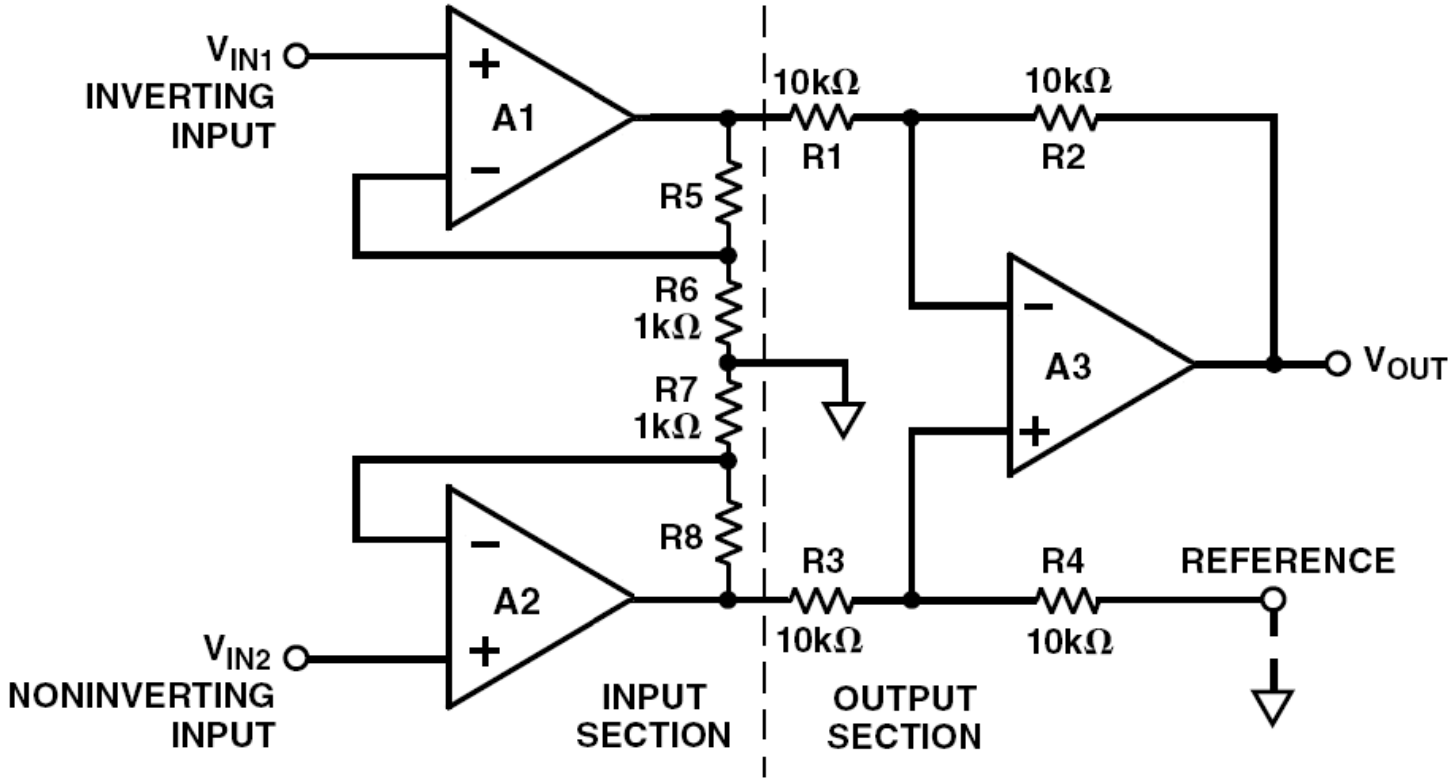


$$V_{OUT} = (V_{IN2} - V_{IN1})\left(\frac{R2}{R1}\right)$$

FOR R1 = R3, R2 = R4

Adapted from Kitchin, C. & Counts, L. (2006)

3 Op Amp In Amp with Gain

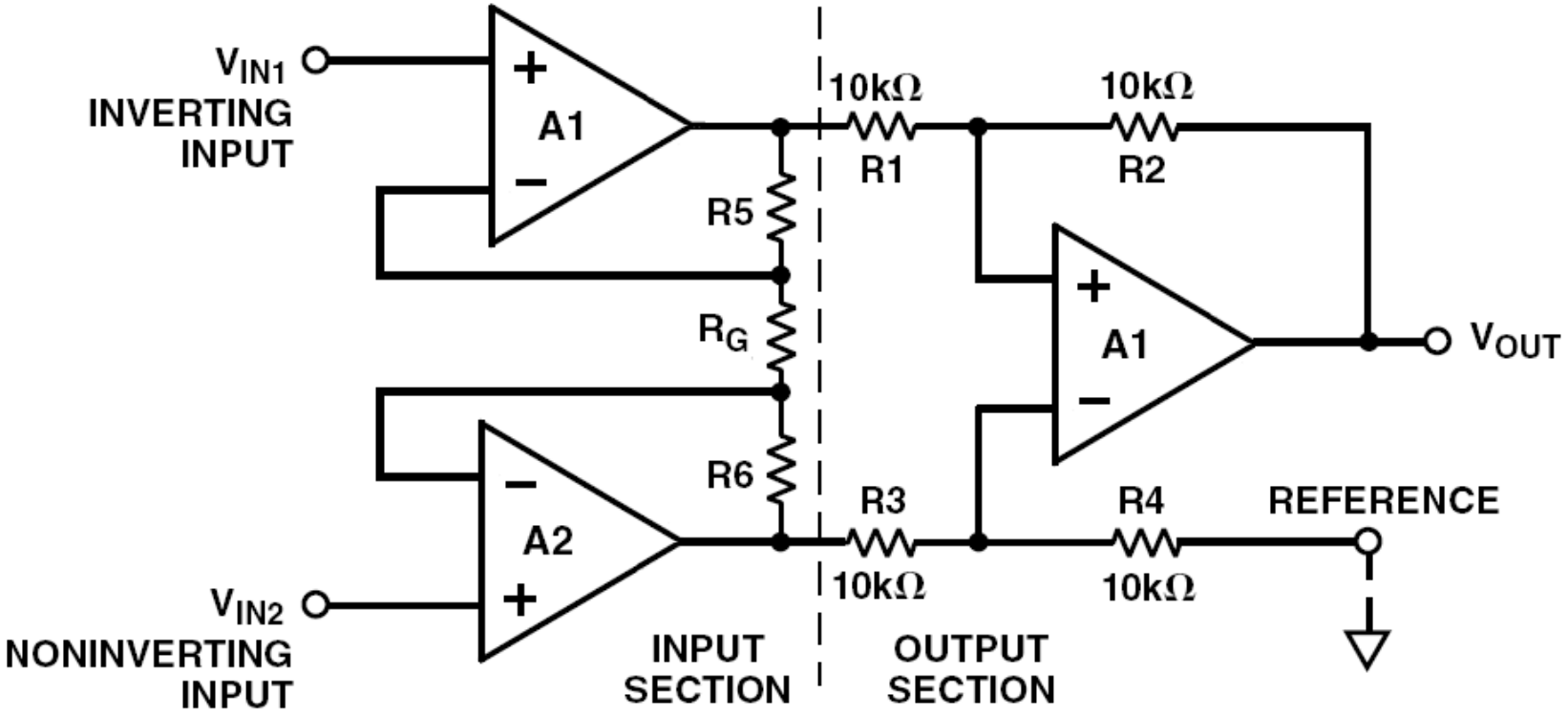


$$V_{OUT} = (V_{IN2} - V_{IN1})\left(1 + \frac{R_5}{R_6}\right)\left(\frac{R2}{R1}\right)$$

FOR R1 = R3, R2 = R4, R5 = R8, R6 = R7

Adapted from Kitchin, C. & Counts, L. (2006)

Classic 3-Op Amp In Amp



$$V_{OUT} = (V_{IN2} - V_{IN1})\left(1 + \frac{2R_5}{R_G}\right)\left(\frac{R2}{R1}\right)$$

FOR R1 = R3, R2 = R4, R5 = R6

Points to Remember

- 3 opamp in-amps can be built out of:
  - Bipolar input opamps
  - FET input opamps
- FET input opamps have:
  - Low bias currents
  - Lower CMR, higher offset voltage and offset drift than Bipolar input opamps
- Good precision opamps often work well (OP07)